HIMANSHU GAUTAM | 24EC10087

B.Tech. (Hons.) in Electronics and Electrical Communication Engineering

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EDUCATION

2028 | B.TECH | IIT Kharagpur — CGPA: 8.24 / 10
2024 | CLASS XII | Kanha International Academy, Etawah — 92%
2022 | CLASS X | St. Francis Academy, Auraiya — 96%

PROJECTS

CareConnect – Healthcare Appointment Platform | *Self Project* — *May'26 – June'26*

- Developed a full-stack MERN platform featuring role-based access control (RBAC) to deliver secure, dedicated dashboards for 3 user roles.
- Built 25+ RESTful APIs using JWT and Express middleware to handle high-concurrency appointment scheduling and profile management.
- Integrated the Razorpay payment gateway for secure transactional workflows, triggering automated real-time booking confirmations.
- Designed user interfaces for doctor availability, appointment scheduling and profile management, booking history through user side.
- Modularized backend into reusable components using Express Router and Mongoose, optimizing codebase scalability and maintainability.

NoteVault – Secure Notes Management | *Self Project* — *Sep'25 – Oct'25*

- Implemented secure authentication using JWT, bcrypt, and 4+ protected API endpoints, ensuring isolated access to user-specific notes.
- Built 8+ RESTful APIs with Express Router and error-handling Middleware ensuring reliable execution of authenticated CRUD workflows.
- Implemented dynamic keyword-based search and editing features, enabling efficient retrieval and management of user-specific notes.
- Optimized frontend-backend communication using Axios and React Router, delivering fast authenticated navigation and API interactions.
- Designed robust Mongoose schemas with validation rules for 2 MongoDB collections, supporting secure storage of user and note data.

CineVerse – Movie Discovery Platform | *Self Project* — *May'25 – Jun'25*

- Developed full-stack MERN movie discovery platform using React.js, integrating 5+ TMDB endpoints to deliver real-time movie exploration.
- Constructed modular data-fetching pipelines to seamlessly format, and serve multi-layered data directly from third-party TMDB APIs.
- Rendered real-time movie data including IMDb ratings, cast details, trailers, and release information through a responsive React interface.
- Implemented a 700 ms debounced live-search with Appwrite-powered search analytics, dynamically ranking the Top 5 trending searches.
- Optimized movie discovery with Axios-powered infinite scrolling, loading 20 movies per request while ensuring client-side data fetching.

PathPilot – Pathfinding Visualizer | *Self Project* — *May'26 – Jun'26*

- Developed an interactive pathfinding visualizer implementing BFS, DFS, and Dijkstra's algorithm on a dynamic grid with animated traversal.
- Implemented Recursive Division Maze Generation and dynamic obstacle creation, enabling users to visualize pathfinding across mazes.
- Visualized visited nodes, shortest paths, and configurable animation speeds on an interactive grid for intuitive algorithm comparison.
- Engineered a custom Min-Heap Priority Queue and path reconstruction logic to efficiently support weighted graph traversal algorithms.

POSITIONS OF RESPONSIBILITY

Tech Subhead | *Kshitij* | *Technology Students' Gymkhana, IIT Kharagpur* — *Aug'25 – Jan'26*

- Co-developed the official Kshitij 2026 website (90K+ visits) and Campus Ambassador Portal using MERN, Three.js, React 3 Fiber, GSAP.
- Coordinated technical operations for Overnite and Source Code, enabling seamless participation for 4,000+ attendees nationwide.

Web and Core Team Member | *Technology Adventure Society (TAdS), IIT Kharagpur* — *Sep'25 – Mar'26*

- Led 20-30 volunteers to execute a flagship event during Shaurya Sports Fest, managing logistics and operations for 20,000+ attendees.
- Oversaw registration, technical support, and on-ground coordination, ensuring smooth execution under high participant volume.

SKILLS AND EXPERTISE

Skills: Competitive Programming | Data Structures and Algorithms | Object Oriented Programming | Web Development
Programming Languages: JavaScript | TypeScript | Python | C | C++ | HTML5 | CSS3 | MySQL
Libraries/Frameworks: C++ STL (Standard Template Library) | React.js | Node.js | Express.js | EJS | BootStrap | Tailwind CSS | GSAP | JWT | Axios | Bcrypt
Softwares/Tools: Git & GitHub | MongoDB | Mongoose | JSON | Postman | Vercel | Netlify | Claudinary | DaVinci Resolve | Blender | Adobe Photoshop | Visual Studio Code

COURSEWORK INFORMATION

Mathematical Courses: Programming and data structures* | Probability and Stochastic | Advanced Calculus | Linear Algebra (Numerical and Complex Analysis) | Optimization Models
Online Courses: Full Stack Web Development (Udemy) | Introduction to Artificial Intelligence and Machine Learning | Data Structures and Algorithms (Striver)
Departmental Courses: Digital Signal Processing* | Digital Electronic Circuits* | Analog Electronic Circuits* | Network Theory* | Electromagnetic Engineering | Signals and Systems | Electrical Technology* | Communications I* | Systems and Control
*Courses marked * have lab component as well*

AWARDS AND ACHIEVEMENTS

- Secured global rank among Top 10% out of 28k+ participants in Codeforces Round 1098 (Div 2) and 14% out of 30k+ in Round 1105 (Div 2).
- Solved 300+ problems on LeetCode, Codeforces, Coding Ninjas, and other focusing on highly optimized data structures and algorithms.
- Earned the Pupil rank on Codeforces (Peak Rating: 1219, Handle: him1617) by consistently solving algorithmic challenges in rated contests.

EXTRA CURRICULAR ACTIVITIES

- Managed event registrations and logistics within the Kshitij Core Team, while executing campus outreach strategies as a Campus Affiliate.
- Earned an EX (Excellent) grade in the first-year NCA Fine Arts program and showcased original artworks in the institute's annual exhibition.
- Collaborated within the Rajendra Prasad Hall teams to engineer large-scale Rangoli and Illumination cultural festival installations.